

Contact

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Top Skills

Java
Visual Studio
C++

Languages

German (Professional Working)
Korean (Limited Working)
Urdu (Native or Bilingual)
English (Full Professional)

Honors-Awards

Silver Medal
Member of Dean's List of Honor
Won science day competition
Final Year Project got funded by the ICT

Publications

TEMPORALFC: A Temporal Fact Checking approach over Knowledge Graphs
DRAZ: SPARQL Query Engine for Heterogeneous Metadata Sources
HybridFC: A Hybrid Fact-Checking Approach for Knowledge Graphs
An Empirical Evaluation of Cost-based Federated SPARQL Query Processing Engines
ExPrompt: Augmenting Prompts Using Examples as Modern Baseline for Stance Classification

Umair Qudus

Marie-Curie fellow and PhD candidate at Paderborn University.
Paderborn, North Rhine-Westphalia, Germany

Summary

I am a computer scientist. I graduated in Computer Science from the National University of Computing and Emerging Sciences, with a strong background in data science, semantic web technologies, and query processing. I worked as a research engineer at Kyung Hee University, South Korea. I am currently working as a Ph.D. candidate at Paderborn University, Germany, tackling the application of Knowledge Graphs such as Fact Checking using traversal, embedding, and NLP techniques to assess and improve the quality of large-scale knowledge graphs.

#FactChecking #KnowledgeGraphs #QueryOptimization
#FederatedQueryProcessing

Experience

Paderborn University
Researcher (Wissenschaftlicher Mitarbeiter)
August 2020 - Present (4 years 1 month)
Paderborn, North Rhine-Westphalia, Germany

Research Fellow at PADERBORN UNIVERSITY, as part of the KnowGraphs project from EU's Horizon 2020. Member of the Data Science (DICE) research group, investigating complex topics such as fact-checking for large-scale knowledge graphs by applying natural language processing and machine learning techniques.

Kyung Hee University
Research Engineer
August 2017 - August 2020 (3 years 1 month)
Suwon, Gyeonggi-do, Korea

Few Research Projects that I have worked on are as following:

1) DataSet Search based on DCAT standards:

By using ontology and search threshold value we created an algorithm to recommend similar datasets based on user input dataset. The brief overview of techniques is as following:

- a. We used machine learning to train our model for recommendation.
- b. We proposed a similarity algorithm using ontology and trained model to calculate the similarity between datasets.
- c. We used datahub datasets (SPARQL endpoints, data dumps and void description) for training our model.
- d. We used DBpedia spotlight tool for extraction of entities and creating ontology for recommendation.
- e. We created a crawler to scan through all datasets and extract required information.

we will soon update our project on our github page.

2) Mapping from SPARQL query to API call:

We created a mapping tool to generate API call from SPARQL query and extract the data from API and reformat the returned JSON data in the form of RDF. Project is available on our github page.

3) Federated Query processing engines quality metric:

This project is for measuring the quality of federated query engines. It is also available on our github page.

Shaheed Zulfikar Ali Bhutto Institute of Science and Technology University Lecturer

January 2017 - August 2017 (8 months)

Islamabad

taught different programming and computer science related courses such as:

1- Machine Learning

2- AI

3- Algorithms

Frontier Works Organization, Pakistan Assistant Manager (Sr Software Engineer) November 2015 - December 2016 (1 year 2 months) Northern Punjab Rawalpindi, Pakistan

Android Application development for FWO

IT Art Technologies Co Founder October 2015 - May 2016 (8 months)

Islamabad

We were a small startup company, Worked on Android and IOS Apps for freelancing clients and our own products. There are few Apps that we have developed and available on Play as well as Amazon App stores.

website: www.itarttech.com

Convo.com

Software Engineer

August 2013 - November 2015 (2 years 4 months)

Islamabad

Worked on Android application for Convo in a team.

Worked on and deployed many features like chat, news feed , Search etc.

About Company:

Convo is simply a facebook for companies. Convo already helps power 6,000 businesses including TechCrunch, CBS, Jones Lang LaSalle, The Next Web, and many others. Convo is proudly backed by Morgenthaler Ventures, the same early-stage VCs that funded Apple, Siri, Lending Club, and Evernote.

National University of Computer and Emerging Sciences

Lab Instructor

January 2011 - June 2013 (2 years 6 months)

H11-4 Islamabad

Conducted Labs for Databases and Android Application Development.

EnMedical Systems

Contractor Software Engineer

June 2012 - September 2012 (4 months)

Islamabad

Worked on and developed a medical software application in a team of 4. Tasks that I worked on and completed:

- Developed application pages related to CRUD (create, update and delete) operations on database.
- Developed and linked signature panel to those pages.
- UI design. Tools used: Wave maker, MYSQL, SVN, Tight VNC Languages used: JAVA, html/JavaScript

Smarnovative Labs

Internee(Mobile Apps developement)

August 2011 - May 2012 (10 months)

Islamabad

I was assigned tasks related to the application development on Android and iPhone in which I worked on the following designs:

- Querying the Databases
- Working on stack mob
- Working on UI design
- Working related to RSS feeds

Advance Computing and Engineering Solution (pvt) Ltd. (ACES),
ISLAMABAD, PAKISTAN.

Internee

June 2011 - August 2011 (3 months)

Islamabad, Pakistan

We, a group of two people, were given a task to develop a shopping website. The development platform used was Microsoft Visual Studio and technology used was ASP.net. We performed following tasks:

- Designed the interface of shopping cart pages
- Coded the functionality behind those pages
- Developed a database in oracle 10g and connected it with the web pages
- Learnt and added PayPal transaction service in our website

Education

Paderborn University

Doctor's Degree, Computer Science · (August 2020 - August 2023)

National University of Computer and Emerging Sciences

Masters of Science (MS), Computer Science · (2014 - 2016)

National University of Computer and Emerging Sciences

Bachelor of Science (BS), Computer Science · (2009 - 2013)

Army public college Rawalpindi

Intermediate, Pre-Engineering · (2007 - 2009)